

BREATHEBOX · REQUIREMENTS

BreatheBox requirements

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	BBX-REQ-001	Version	0.9
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/breathebox		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.9	2026-10-02	Amish Chadha	Approved follow-ups carried out (BBX-CAL-001 v0.7): sash jammer now BOM line 14 (R12); R9 mass 11.8 kg with the longer struts and the sash jammer; R15 USD 294, USD 39 over the value-engineering target; no status category and no target changed
0.8	2026-10-02	Amish Chadha	R12: sash held by a bought no-drill sash jammer, decided by Amish on 2026-10-02 (BBX-DEC-001)
0.7	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.6	2026-09-30	Amish Chadha	Status against the constructable design (BBX-DDR-003, BBX-CAL-001 v0.4); R15 Met to Not met at \$283; R3, R9 and R10 figures updated; no target changed
0.5	2026-09-26	Amish Chadha	Budget approved by Amish; R15 target \$255, status Not met to Met
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)

0.3	2026-09-25	Amish Chadha	Status against BBX-CAL-001 at TRL 3; R7 condensate figure corrected to 0.21 L/h; R8 and R15 notes per BBX-DDR-001 (adopted for TRL 3, open for Amish's review)
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

BreatheBox requirements

These requirements are proposals for review, not yet validated with users, and will be revised after co-design (see BBX-PRB-001). Status is judged against the TRL 3 calculations in BBX-CAL-001 and the parametric model in `cad/src/model.py` ; nothing has been built or measured. No target was relaxed at TRL 3. On 2026-09-25 Amish accepted the review recommendations (BBX-DDR-002): R15 read against a \$250 value-engineering target (it was \$220), and R4 and R6 show the quiet night mode; R6 itself is unchanged until a real fan datasheet has been checked. The R7 condensate figure was corrected from 0.15 to 0.21 L/h because the calculation showed a higher worst case. The value-engineering target was set at \$255 on 2026-09-26, and the concept's priced BOM was within it. On 2026-09-30 the design was made constructable (BBX-DDR-003): the parts it needed brought the BOM to \$283. On 2026-10-02 Amish's decisions added a bought sash jammer and a lower wall foot with longer struts. Value-engineering target: USD 255. Estimated cost of the constructable design: USD 294 (USD 39 over the target); BBX-DEC-001 lists the savings worth trying. No target changed.

The **design case** is a 30 m³ bedroom with two sleeping adults, 20 °C and 50 % relative humidity indoors, 0 °C outdoors, and a vertical sliding sash window with a 900 mm clear width.

Table 1. Requirements and status at TRL 3.

ID	Requirement	Target	Status at TRL 3 (BBX-CAL-001)	Verification (TRL 4 or later)
R1	Balanced ventilation over a useful range	Supply and exhaust each adjustable from 30 to 70 m ³ /h (18 to 41 cfm); supply and exhaust within 10 % of each other outside frost mode	Met on paper: 80 m ³ /h per stream at full speed with loaded filters; balance by fan tach	Flow hood measurement
R2	Recover heat from the exhaust air	Sensible effectiveness 75 % or more at 50 m ³ /h and balanced flow	Met on paper: 79.8 %	Temperatures on the four ports
R3	Low running power	Fans and controls 12 W or less at 50 m ³ /h	Met on paper: 5.7 W clean, 7.5 W with loaded filters	Power meter
R4	Keep bedroom air fresh	CO ₂ at or below 1,000 ppm (24-hour average) in the design case, following the Health Canada residential guideline	Met on paper: 607 ppm 24 h mean at 50 m ³ /h; 980 ppm overnight. In night mode (32 m ³ /h): 712 ppm 24 h mean, about 1,300 ppm overnight	CO ₂ logging
R5	Filter the supply air	ISO 16890 ePM1 50 % or better on supply; coarse filter on exhaust to protect the core	Met by specification	Filter data sheet
R6	Quiet at night	30 dB(A) or less at 1 m from the room face at 50 m ³ /h	Not met at 50 m ³ /h: about 39 dB(A) (estimate). Night mode at 32 m ³ /h (BBX-DDR-002): about 30 dB(A) clean, 32 dB(A) with loaded filters	Sound meter
R7	Handle condensate and frost	All condensate drains outdoors (up to about 0.21 L/h); runs without core blockage down to -10 °C outdoor, using a frost mode that may reduce supply flow	Met on paper: frost onset about -2.2 °C; at -10 °C the supply runs at 33 % of the exhaust	Cold-chamber or winter test
R8	Fit common windows	Vertical sliding sash windows with 700 to 1,000 mm clear width and sill 150 mm or deeper; adapter panel for horizontal sliders	Met on paper for sash: the hoods need 686 mm. Casement and tilt-and-turn windows are a later variant (BBX-DDR-001 item A2, decided by Amish in BBX-DDR-002)	Fit check against a window survey
R9	Renter-friendly installation	No drilling of frame or wall; one person installs or removes in 30 min or less; unit mass 12 kg or less	Not verifiable at TRL 3: mass 11.8 kg met on paper (sash jammer included; margin 0.2 kg); install time needs a trial	Timed trials
R10	Electrical safety	Only 24 V SELV inside the unit, from a certified plug-in adapter; fused low-voltage input; no user-accessible mains	Met by design: 36 W adapter, 2 A fuse; worst case 35.4 W	Design review
R11	Weather and outdoor air	Rain-proof downward-facing hoods; insect mesh 1.5 mm or finer; 400 mm or more between supply intake and exhaust outlet	Met on paper: 420 mm; 1 mm mesh	Spray test

ID	Requirement	Target	Status at TRL 3 (BBX-CAL-001)	Verification (TRL 4 or later)
R12	Security and child safety	Raised sash lockable onto the insert panel; no opening wider than 100 mm through the installed unit	Met by design, unverified: largest opening 2.35 mm; the raised sash is held down on the panel by a bought no-drill sash jammer in the inner track (decided 2026-10-02; bill of materials line 14)	Design review with the sash jammer fitted
R13	Serviceable	Filters changed without tools in 2 min or less; core removable for washing; all parts replaceable with a screwdriver	Not verifiable at TRL 3: lift-off lid and slide-out filters and core in the model	Timed trials
R14	Local data only	CO ₂ , humidity and temperature shown on the unit; works with no network or cloud account; optional local logging	Not verifiable at TRL 3: no firmware yet	Firmware review
R15	Affordable	Prototype parts at or under the \$255 value-engineering target (a hypothetical control target; was \$250, and \$220 before that; BBX-DDR-002)	Over the value-engineering target by \$39: \$294 (priced BOM of the constructable design, BBX-DDR-003, with the sash jammer)	Priced BOM (bom/bom.csv)

■ Requirements not met

- **R6 (noise) not met:** about 39 dB(A) at 1 m at 50 m³/h against 30 dB(A). The accepted night mode (32 m³/h) gives about 30 dB(A) with clean filters and 32 dB(A) with loaded filters; larger, slower fans or a silencer are to be checked against a real fan datasheet (on hold with TRL 4) before any change to R6.
- **R15 (cost) over the value-engineering target by \$39:** Value-engineering target: USD 255. Estimated cost of the constructable design: USD 294 (USD 39 over the target). The target is `budget_usd` in `project.yaml`, set on 2026-09-26 (BBX-DDR-002). The constructable design added partitions, filter seats, grilles, latches, flanges, bracket parts and bolts that the concept needed but did not list (BBX-DDR-003); the 2026-10-02 decisions added the sash jammer (\$10) and longer struts (\$1). The savings worth trying are in the design decisions register (BBX-DEC-001).
- **Not verifiable at TRL 3:** R9 (install time), R13 (filter change time) and R14 (firmware).

■ Assumptions

- Air density 1.2 kg/m³ and specific heat 1,005 J/(kg·K).
- CO₂ generation about 0.014 m³/h per sleeping adult (estimate); outdoor CO₂ about 420 ppm.
- Core, filter, fan and noise values are class values stated in BBX-CAL-001 section 1, not datasheet values.